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GRADE A+**
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Operational Sales Performance Dashboard

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Operational Sales Performance Dashboard

Github Repository Link: <https://github.com/LikithaSrinivas100/Business-Analytics>

Problem Statement

An enterprise retail organization relies heavily on manually prepared standalone spreadsheets to track and audit ongoing business performance. This disconnected workflow leads to high data latency, version control errors, and cross-departmental alignment gaps across primary commercial boundaries.

To streamline tracking, the data analyst must build a centralized operational dashboard in Power BI Desktop. The reporting asset must consolidate distributed transactional files, map unified underlying relation schemas, compute standardized high-level business indicators via DAX measures, and extract interactive cross-filtering paths over customer segments, temporal demand vectors, and spatial categories to drive real-time operational decisions.

Methodology / Steps Followed

Step 1: Data Import

- Opened Power BI Desktop.
- Imported the retail company dataset using the **Get Data** option.
- Loaded the data tables using **Import Mode** to ensure high performance.

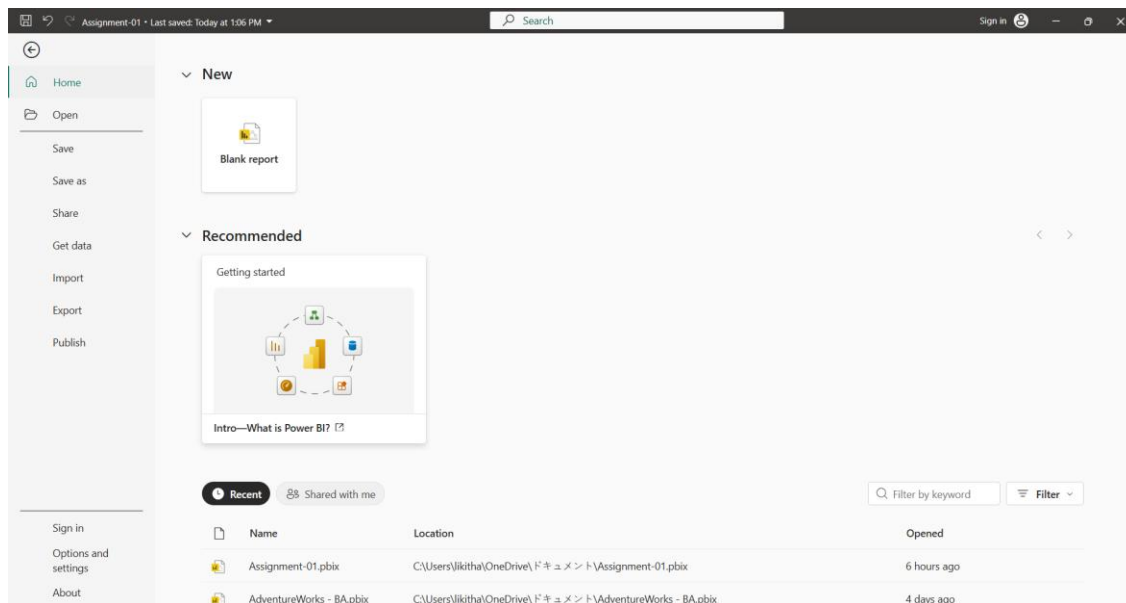


Figure 1: Data Import Interface

Step 2: Data Cleaning

- Opened **Power Query Editor** to clean the raw data.
- Checked and adjusted column data types (Text for categories/IDs, Decimals for revenue and metrics).
- Removed blank rows and filtered out invalid records.

Step 3: Data Modeling

- Opened the **Model View** tab.
- Created relationships between the Sales transaction table and dimension tables (Products, Customers, and Regions).
- Verified that the model formed a clean **Star Schema** with one-to-many relationships.

Step 4: Dashboard Development

- Dragged and dropped visual elements onto the canvas to construct the report.
- Added **KPI cards** to display high-level totals.

- Built a **Line chart** to show monthly trend lines, a **Column chart** for product line metrics, and a **Donut chart** for regional segmentation.
- Configured dynamic **Slicers** at the top for interactive filtering.

Output

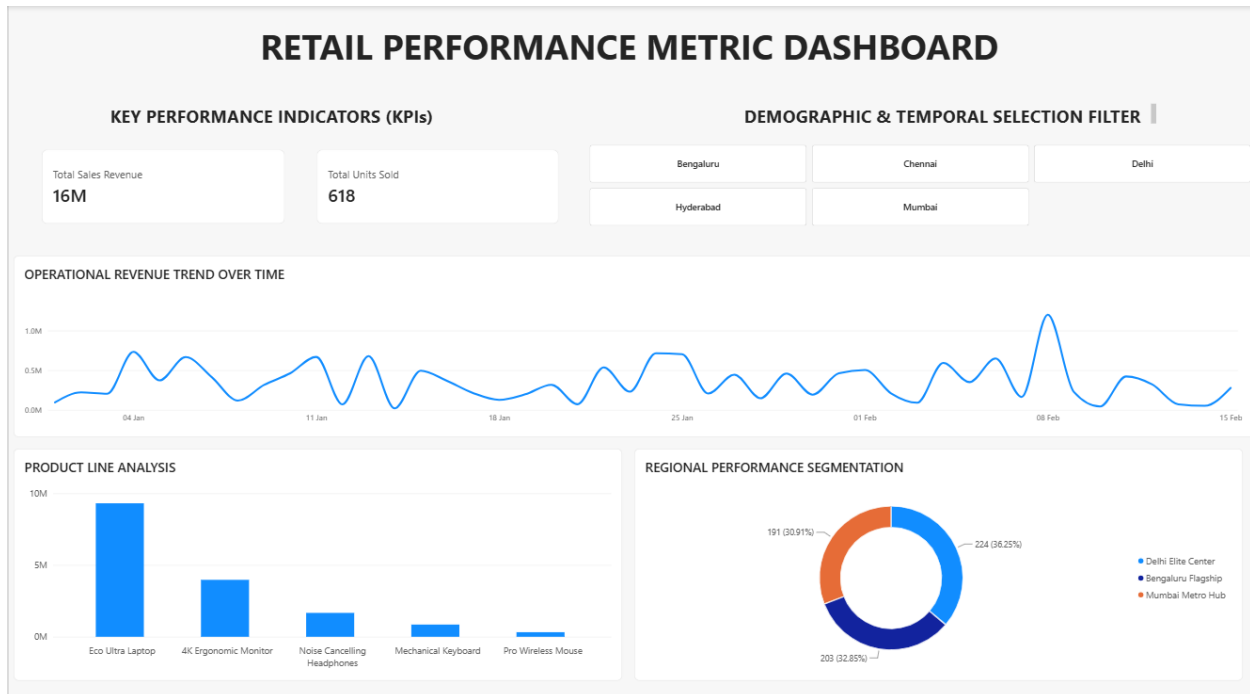


Figure 2: Retail Performance Metric Dashboard

The completed interactive dashboard displays the following operational metrics:

- **Key Performance Indicators:** Shows Total Sales Revenue (16M) and Total Units Sold (618) on clear top cards.
- **Operational Revenue Trend Over Time:** A wide line graph tracking sales fluctuations over dates.
- **Product Line Analysis:** A column chart showing that the Eco Ultra Laptop is the top revenue-generating item.

- **Regional Performance Segmentation:** A donut chart showing sales distribution across branches: Delhi Elite Center (36.25%), Bengaluru Flagship (32.85%), and Mumbai Metro Hub (30.91%).
- **Selection Filters:** Interactive city buttons (Bengaluru, Chennai, Delhi, Hyderabad, Mumbai) that filter the entire page instantly.

Business Insights

- **Top Product Driver:** The business relies heavily on high-value items, as the Eco Ultra Laptop drives the vast majority of sales revenue.
- **Balanced Regional Performance:** Sales workloads are evenly split across territories, with Delhi slightly leading at 36.25%.
- **Trend Volatility:** The revenue line chart shows sharp spikes over time, indicating bulk business purchases rather than steady daily retail transactions.

Conclusion

This project successfully replaced the company's manual reporting process with an automated Power BI dashboard. By importing, cleaning, and modeling the data, raw transactional information was turned into a centralized visual tracking tool that helps management make fast, data-driven business choices.

Learning Outcomes

Through this project, I learned:

- How to import and clean business datasets using Power Query.
- How to establish relationships to build a functional data model.
- How to design interactive dashboards with KPI cards, line charts, column charts, and donut charts.
- How to use global slicers for dynamic filtering.
- How to generate actionable business insights from a visual dashboard layout.